Database Content

The database consists of several sets of data: food descriptions, nutrients, weights and measures, footnotes, and sources of data. The sections below provide details about the information in each.

Food Descriptions

This file includes descriptive information about the food items. For more details on the format of the Food Description file, see “Food Description File Formats” (p. 29). A full description (containing the name of the food with relevant characteristics, e.g., raw or cooked, enriched, color) and a short description (containing abbreviations) are provided. Abbreviations used in creating short descriptions are given in Appendix A. In creating the short description, the first word in the long description is not abbreviated. In addition, if the long description is 25 characters or less, the short description contains no abbreviations. Abbreviations used elsewhere are given in Appendix B. Brand names used in food descriptions are in upper case. Scientific names, common names, manufacturers’ names, amounts of refuse, and refuse descriptions are provided where appropriate. The common name field includes alternative names for a product, e.g., soda or pop, for a carbonated beverage. In addition this field also includes Uniform Retail Meat Identity Standard (URMIS) identification numbers and USDA commodity codes as appropriate. The food group to which the food item belongs is also indicated. A code is also provided indicating if the item is used in the Food and Nutrient Database for Dietary Studies (FNDDS; USDA, ARS, 2012). The factors used to calculate protein from nitrogen are included, as well as those used to calculate kilocalories. There are no factors for items prepared using the recipe program of the NDBS or for items where the manufacturer calculates protein and kilocalories.

The refuse and refuse description fields contain amounts and descriptions of inedible material (for example, seeds, bone, and skin) for applicable foods. These amounts are expressed as a percentage of the total weight of the item as purchased, and they are used to compute the weight of the edible portion. Refuse data were obtained from NFNAP and other USDA-sponsored contracts and US Department of Agriculture Handbooks 102 (Matthews and Garrison, 1975) and 456 (Adams, 1975). To calculate the “amount of nutrient in edible portion of 1 pound (453.6 grams) as purchased,” use the following formula:

$$Y = V*4.536*[(100-R)/100]$$

where:

Y = nutrient value per 1 pound as purchased,
V = nutrient value per 100 g (Nutr_Val in the Nutrient Data file), and
R = percent refuse (Refuse in the Food Description file).

For meat cuts containing bone and connective tissue, the amount of connective tissue is included in the value given for bone. Separable fat is not shown as refuse if the meat is described as separable lean and fat. Separable fat generally refers to seam fat and external trim fat. Separable lean refers to muscle tissue that can be readily separated from fat, bone, and connective tissue in the intact cut; it includes any fat striations (marbling) within the muscle. For boneless cuts, the refuse value applies to connective tissue or connective tissue plus separable fat. The percentage yield of cooked, edible meat from 1 pound of raw meat with refuse can be determined by using the following formula:

$$Y = (W_c/453.6)*100$$

where:

Y = percentage yield of cooked edible meat per 1 pound as purchased, and

W_c = weight of cooked, edible meat in grams.

Food Group. To facilitate data retrieval the food items in SR are organized into food groups. Currently there are 25 food groups, which are listed in the Food Group Description file. For more details on the format of this file, see “Food Group Data File Formats” (p. 31). Starting with SR25, the food group, “Ethnic Foods” has been renamed “American Indian/Alaska Native Foods” to better reflect its contents. Data on other ethnic foods are contained in their respective food groups, for example data on plantains, a Latino ethnic food are in food group 9 (Fruit and Fruit Juices), while the Asian foods, miso and natto, are entered in food group 16 (Legumes and Legume Products). Food group 36 (Restaurant Foods) contains foods obtained from casual dining, full service restaurants, Latino restaurants, and Chinese restaurants (not Fast Foods, which are in food group 21). Home prepared items or prepared frozen entrees are included in Food Group 22, Meals, Entrees, and Side dishes. Some food items, such as beverages and rice, though obtained at restaurants are included in their respective food groups.

LanguaL. To address the needs of diverse users of the USDA food composition databases, starting with SR23, NDL is providing an expanded standardized food description for selected food groups (spices and herbs, fruits and fruit juices, pork products, vegetables and vegetable products, and beef products) based on the LanguaL Thesaurus (Moeller and Ireland, 2009). The use of this multi-hierarchical food classification system will permit the harmonization of food description terms and definitions across many cultures and languages to support food research, food safety, nutrition monitoring, and food marketing**.**

LanguaL stands for "**Langua aL**imentaria" or "language of food". Work on LanguaL was started in the late 1970's by the Center for Food Safety and Applied Nutrition (CFSAN) of the United States Food and Drug Administration as a co-operative effort of specialists in food technology, information science, and nutrition.

Since then, LanguaL has developed in collaboration with the NCI, and, more recently, its European partners, notably France, Denmark, Switzerland, and Hungary. Since 1996, the European LanguaL Technical Committee has administered the thesaurus.

The thesaurus provides a standardized language for describing foods, specifically for classifying food products for information retrieval. LanguaL is based on the concept that:

- Any food (or food product) can be systematically described by a combination of characteristics or facets;
- These characteristics can be categorized into viewpoints and coded for computer processing; and
- The resulting viewpoint/characteristic codes can be used to retrieve data about the food from external databases.

The current facets for foods in SR27 include: product type; food source; part of plant or animal; physical state, shape or form; extent of heat treatment; cooking method; treatment applied; preservation method; packing medium; container or wrapping; food contact surface; consumer group/dietary use/label claim; geographic places and regions; and adjunct characteristics of food.

The specific tables added to SR are the LanguaL Factor File (p. 31) and the LanguaL Factors Description File (p. 31). For more information on LanguaL, see the web site: http://www.langual.org.

Nutrients

The Nutrient Data file contains mean nutrient values per 100 g of the edible portion of food, along with fields to further describe the mean value. The following statistical attributes are provided to better describe the data:

- Nutrient value—the mean of the data values for a specific parameter. Nutrient values have been rounded to the number of decimal places for each nutrient as specified in the Nutrient Definition file (p. 34).
- Number of data points—the number of data points used to estimate the mean.
- Standard error—the standard error of the mean: a measure of variability of the mean value as a function of the number of data points.
- Number of studies—the number of analytical studies used to generate the mean. A study is a discrete research project conducted or reported for a specific food. A study can be the analysis of one nutrient in one food, one nutrient in many foods, or many nutrients in many foods.
- Minimum value—the smallest observed value in the range of values.
- Maximum value—the largest observed value in the range of values.
- Degrees of freedom—the number of data values that are free to vary after certain restrictions are placed on the estimates; used in probability calculations.

- Lower- and upper-error bounds—represents a range of values within which the population mean is expected to fall, given a pre-specified confidence level. For SR27 and related releases, the confidence level is 95 percent.
- Statistical comments—provide additional details about certain assumptions made during statistical calculations. The definition of each comment is given after the description of the Nutrient Data file under "File Formats" (p. 31).

Other fields provide information on how the values were generated, as follows:

- Source code—contains codes indicating the type of data (e.g., analytical, calculated, assumed zero) in the Nutrient Data file.
- Derivation code—gives more information about how a value was calculated or imputed. Procedures used to impute a nutrient value are described by Schakel *et al*. (1997).
- Reference NDB number—indicates the NDB number of the food item that was used to impute a nutrient value for another food. This field is only populated for items which have been added or updated since SR14 for which an imputed value is provided.
- Added nutrient marker—a "Y" indicates that a mineral or vitamin was added for enrichment or fortification. This field is populated for ready-to-eat breakfast cereals and many brand-name hot cereals in food group 08. In future releases, this field will be populated for other food groups, where applicable.
- AddMod_Date—indicates when a value was either added to the database or last modified. This field was first added with SR24 (2011). Data, which have not been updated since SR14 (2001) carry the date when that section of AH-8 was published. When the nutrient values are reviewed, but not modified, there is no change in the AddMod_Date. Hence, the field may not accurately reflect the currency of the data. Dates associated with calculated values (carbohydrate by difference, energy, vitamin A (IU and RAE), and folate DFE) may carry newer dates reflecting their recalculation when other values in the profile of a particular food item were updated, though the values from which the specific value was calculated may not have changed. Only values added or modified since SR14 will have newer dates. To determine if the date has changed the values between the current and preceding releases are compared to the number of decimal places specified in the Nutr_Def table. The date associated with the source documents used to determine the mean can be found in the Sources of Data file. The description of this file can be found on p. 37.
- Confidence code—indicates the relative quality of the data. This code is derived using the data quality criteria first described by Mangels *et al*. (1993). These criteria have been expanded and enhanced for the NDBS (Holden *et al*., 2002). This field is included as a placeholder for future releases.

For more details on the Nutrient Data file, see "Nutrient Data File Formats" (p. 32). Nutrient values indicate the total amount of the nutrient present in the edible portion of the food, including any nutrients added in processing. Table 1 gives an idea of the

comprehensiveness of the database by listing for each nutrient the number of food items that contain data.

Table 1.—Number of Foods in the Database (*n* = 8618) Containing a Value for the Specified Nutrient

Nutr. No	Nutrient	Count
255	Water *†	8612
208	Energy *†	8618
268	Energy	8584
203	Protein *†	8618
204	Total lipid (fat) *†	8618
207	Ash *†	8286
205	Carbohydrate, by difference *†	8618
291	Fiber, total dietary *†	7962
269	Sugars, total *†	6679
210	Sucrose	1544
211	Glucose (dextrose)	1551
212	Fructose	1550
213	Lactose	1530
214	Maltose	1518
287	Galactose	1396
209	Starch	1041
301	Calcium, Ca *†	8264
303	Iron, Fe *†	8471
304	Magnesium, Mg *†	7936
305	Phosphorus, P *†	8046
306	Potassium, K *†	8208
307	Sodium, Na *†	8535
309	Zinc, Zn *†	7917
312	Copper, Cu *†	7363
315	Manganese, Mn †	6478
317	Selenium, Se *†	6868
313	Fluoride, F	532
401	Vitamin C, total ascorbic acid *†	7826
404	Thiamin *†	7939
405	Riboflavin *†	7961
406	Niacin *†	7937
410	Pantothenic acid †	6460
415	Vitamin B-6 *†	7677
417	Folate, total *†	7373
431	Folic acid *†	6631
432	Folate, food *†	6863
435	Folate, DFE *†	6612
421	Choline, total *†	4534
454	Betaine	2075
418	Vitamin B-12 *†	7427
578	Vitamin B-12, added *	4668
320	Vitamin A, RAE *†	7089
319	Retinol *†	6806
321	Carotene, beta *†	5356
322	Carotene, alpha *†	5263
334	Cryptoxanthin, beta *†	5252
318	Vitamin A, IU *†	7932
337	Lycopene †	5228
338	Lutein + zeaxanthin *†	5206
323	Vitamin E (alpha-tocopherol) *†	5613
573	Vitamin E, added *	4501
341	Tocopherol, beta	1763
342	Tocopherol, gamma	1760
343	Tocopherol, delta	1744
344	Tocotrienol, alpha	1353
345	Tocotrienol, beta	1367
346	Tocotrienol, gamma	1357
347	Tocotrienol, delta	1354
328	Vitamin D (D2 + D3) *†	5319
325	Vitamin D2 (ergocalciferol)	76
326	Vitamin D3 (cholecalciferol)	1890

Nutr. No	Nutrient	Count
324	Vitamin D †	5320
430	Vitamin K (phylloquinone) *†	4969
429	Dihydrophylloquinone	1348
428	Menaquinone-4	523
606	Fatty acids, total saturated *†	8274
607	4:0 *	5589
608	6:0 *	5621
609	8:0 *	5836
610	10:0 *	6207
611	12:0 *	6460
696	13:0	263
612	14:0 *	6828
652	15:0	2166
613	16:0 *	7037
653	17:0	2552
614	18:0 *	7027
615	20:0	2628
624	22:0	2101
654	24:0	2016
645	Fatty acids, total monounsaturated *†	7947
625	14:1	2520
697	15:1	1871
626	16:1 undifferentiated *	6799
673	16:1 c	1201
662	16:1 t	1073
687	17:1	2247
617	18:1 undifferentiated *	7053
674	18:1 c	1668
663	18:1 t	1717
859	18:1-11 t (18:1t n-7)	152
628	20:1	6213
630	22:1 undifferentiated *	5676
676	22:1 c	984
664	22:1 t	717
671	24:1 c	1206

Nutr. No	Nutrient	Count
646	Fatty acids, total polyunsaturated *†	7954
618	18:2 undifferentiated *	7070
675	18:2 n-6 c,c	1629
670	18:2 CLAs	1296
669	18:2 t,t	353
666	18:2 i	60
665	18:2 t not further defined	1003
619	18:3 undifferentiated *	6972
851	18:3 n-3 c,c,c (ALA)	1772
685	18:3 n-6 c,c,c	1482
856	18:3i	378
627	18:4	5603
672	20:2 n-6 c,c	2283
689	20:3 undifferentiated	2084
852	20:3 n-3	872
853	20:3 n-6	1073
620	20:4 undifferentiated *	6222
855	20:4 n-6	162
629	20:5 n-3 (EPA) *	5693
857	21:5	117
858	22:4	890
631	22:5 n-3 (DPA) *	5649
621	22:6 n-3 (DHA) *	5651
605	Fatty acids, total trans	3243
693	Fatty acids, total trans-monoenoic	1696
695	Fatty acids, total trans-polyenoic	1400
601	Cholesterol *†	8250
636	Phytosterols	499
638	Stigmasterol	132
639	Campesterol	131
641	Beta-sitosterol	132
501	Tryptophan	5129
502	Threonine	5175

Nutr. No	Nutrient	Count
503	Isoleucine	5179
504	Leucine	5178
505	Lysine	5192
506	Methionine	5191
507	Cystine	5026
508	Phenylalanine	5175
509	Tyrosine	5145
510	Valine	5179
511	Arginine	5165
512	Histidine	5172
513	Alanine	5121
514	Aspartic acid	4966
515	Glutamic acid	5124
516	Glycine	5121
517	Proline	5112
518	Serine	5121
521	Hydroxyproline	1393
221	Alcohol, ethyl *	5345
262	Caffeine *	5143
263	Theobromine*	5103

*Indicates the 65 nutrients included in the USDA Food and Nutrient Database for Dietary Studies (FNDDS).
† Nutrients included in the Abbreviated file (p. 38).

In general, levels of fortified nutrients are the values calculated by the manufacturer or by NDL, based on the Nutrition Labeling and Education Act (NLEA) label declaration of % Daily Value (DV) (CFR, Title 21, Pt. 101) (US Food and Drug Administration–Department of Health and Human Services, 2012). Such values represent the minimum nutrient level expected in the product. If analytical values were used to estimate levels of added nutrients, a number is present in the sample count field for these nutrients.

***Nutrient Retention and Food Yield*.** When nutrient data for prepared or cooked products are unavailable or incomplete, nutrient values are calculated from comparable raw items or by recipe. When values are calculated in a recipe or from the raw item, appropriate nutrient retention (USDA, 2007) and food yield factors (Matthews and Garrison, 1975) are applied to reflect the effects of food preparation on food weights and nutrient content. To obtain the content of nutrient per 100 g of cooked food, the nutrient content per 100 g of raw food is multiplied by the nutrient retention factor and, where appropriate, adjustments are made for fat and moisture gains and losses.

Nutrient retention factors are based on data from USDA research contracts, research reported in the literature, and USDA publications. Most retention factors were calculated by the True Retention Method (%TR) (Murphy *et al*., 1975). This method, as shown below, accounts for the loss or gain of moisture and the loss of nutrients due to heat or other food preparation methods:

$$\%TR = (N_c * G_c) / (N_r * G_r) * 100$$

where:

TR = true retention
N_c = nutrient content per g of cooked food,
G_c = g of cooked food,
N_r = nutrient content per g of raw food, and
G_r = g of food before cooking.

***Proximates*.** The term proximate component refers to those macronutrients that include water (moisture), protein, total lipid (fat), total carbohydrate, and ash. Except in the case of a few foods, nutrient profiles have values for the proximate components and at least one other nutrient.

Protein. The values for protein were calculated from the amount of total nitrogen (N) in the food, using the specific conversion factors recommended by Jones (1941) for most food items. The analytical methods used to determine the nitrogen content of foods are AOAC 968.06 (4.2.04), 992.15 (39.1.16) and 990.03 (combustion) and 991.20 (Kjeldahl) (AOAC, 2010). The specific factor applied to each food item is provided in the N_Factor field in the Food Description file. The general factor of 6.25 is used to calculate protein in items that do not have a specific factor. When the protein content of a multi-ingredient food (e.g., beef stew) is calculated using the recipe program of the NDBS the specific nitrogen to protein conversion factors are applied at the ingredient level. Therefore, the N-factor field will remain empty. When the manufacturer calculates protein the N-factor field will also be empty.

Protein values for chocolate, cocoa, coffee, mushrooms, and yeast were adjusted for non-protein nitrogenous material (Merrill and Watt, 1973). The adjusted protein conversion factors used to calculate protein for these items are as follows:

chocolate and cocoa	4.74
coffee	5.3
mushrooms	4.38
yeast	5.7

When these items are used as ingredients, such as chocolate in chocolate milk or yeast in bread, only their protein nitrogen content was used to determine their contribution to the calculated protein and amino acid content of the food. Protein calculated from total nitrogen, which may contain non-protein nitrogen, was used in determining carbohydrate by difference. This unadjusted protein value is not given in the Nutrient Data file for SR27; rather, it was previously included as a footnote in printed sections of "Agriculture Handbook 8."

For soybeans, nitrogen values were multiplied by a factor of 5.71 (Jones, 1941) to calculate protein. The soybean industry, however, uses 6.25 to calculate protein. To convert these values divide the proteins value by 5.71, and then multiply the resulting value by 6.25. It will also be necessary to adjust the value for carbohydrate by difference (Nutr. No. 205).

Total Lipid. The total lipid (fat) content of most foods obtained through NFNAP is determined by gravimetric methods, including acid hydrolysis (AOAC 922.06, 925.12, 989.05, or 954.02) and extraction methods using a mixed solvent system of chloroform and methanol (AOAC 983.25 or Folch *et al*. 1957). Older values may have been obtained by ether extraction (AOAC 920.39, 933.05, or 960.39). Total lipid determined by extraction is reported as Nutrient No. 204. It is sometimes referred to as "crude fat" and includes the weight of all lipid components, including glycerol, soluble in the solvent system. Nutrient No. 204 may not be identical to the fat level declared on food labels under the NLEA, where fat is expressed as the amount of triglyceride that would produce the analytically determined amount of lipid fatty acids and does not include other lipid components not soluble in the solvent system. The term "NLEA fat" is commonly referred to as "total fatty acids expressed as triglycerides."

Ash. The ash content of foods is determined by gravimeteric methods (AOAC 923.03, 942.05, or 945.46).

Moisture. The moisture (or water) content of foods is determined by vacuum oven (AOAC 934.01, 934.06, 964.22) or forced air (AOAC 950.46).

Carbohydrate. Carbohydrate, when present, is determined as the difference between 100 and the sum of the percentages of water, protein, total lipid (fat), ash, and, when present, alcohol. Total carbohydrate values include total dietary fiber. Available carbohydrate, which is used in some countries, can be calculated if desired by the user, by subtracting the sum of the percentages of water, protein, total lipid (fat), ash, total dietary fiber, and alcohol (when present) from 100. Carbohydrate in beer and wine is determined by methods 979.06 (27.1.21) and 985.10 (28.1.18) of AOAC International (AOAC, 2010), respectively. Total dietary fiber content is determined by enzymatic-gravimetric methods 985.29 or 991.43 of the AOAC (2010). Total sugars is the term used for the sum of the individual monosaccharides (galactose, glucose, and fructose) and disaccharides (sucrose, lactose, and maltose). Analytical data for individual sugars obtained through NFNAP were determined by liquid chromatography (AOAC 982.14). Earlier values were also determined using AOAC methods (2010), with either high-performance liquid chromatography (HPLC) or gas-liquid chromatography (GLC). When analytical data for total sugars are unavailable for items in the FNDDS, values are imputed or obtained from manufacturers and trade associations. Starch is analyzed using the AOAC method 966.11 (2010) or by a polarometric method (The Feedings Stuffs Regulations 1982). Because the analyses of total dietary fiber, total sugars, and starch are performed separately and reflect the analytical variability inherent to the measurement process, the sum of these carbohydrate fractions may not equal the carbohydrate-by-difference value.

Food Energy. Food energy is expressed in kilocalories (kcal) and kilojoules (kJ). One kcal equals 4.184 kJ. The data represent physiologically available energy, which is the energy value remaining after digestive and urinary losses are deducted from gross energy. Energy values, with the exception of multi-ingredient processed foods, are

based on the Atwater system for determining energy values. Derivation of the Atwater calorie factors is discussed in “Agriculture Handbook 74” (Merrill and Watt, 1973). For multi-ingredient processed foods (source codes 8 or 9; for more information on source codes, see p. 34) kilocalorie values generally reflect industry practices (as permitted by NLEA) of calculating kilocalories as 4, 4, or 9 kilocalories per gram of protein, carbohydrate, and fat, respectively, or as 4, 4, or 9 kilocalories per gram of protein, carbohydrate minus insoluble fiber, and fat. The latter method is often used for high-fiber foods.

Calorie factors for protein, fat, and carbohydrates are included in the Food Description file. For foods containing alcohol, a factor of 6.93 is used to calculate kilocalories per gram of alcohol (Merrill and Watt, 1973). No calorie factors are given for items prepared using the recipe program of the NDBS. Instead, total kilocalories for these items equal the sums of the kilocalories contributed by each ingredient after adjustment for changes in yield, as appropriate. For multi-ingredient processed foods, if the kilocalories calculated by the manufacturer are reported, no calorie factors are given.

Calorie factors for fructose and sorbitol, not available in the Atwater system, are derived from the work of Livesay and Marinos (1988). Calorie factors for coffee and tea are estimated from those for seeds and vegetables, respectively.

Minerals. Minerals included in the database are calcium, iron, magnesium, phosphorus, potassium, sodium, zinc, copper, manganese, selenium, and fluoride. Levels of minerals for most foods are determined by methods of the AOAC (2010). Calcium, iron, magnesium, phosphorus, sodium, potassium, zinc, copper, and manganese are usually determined by inductively coupled plasma emission spectrophotometry (AOAC 984.27) or, except for phosphorus, by atomic absorption (AOAC 985.35), with phosphorus determined colorimetrically by AOAC 2.019, 2.095 and 7.098.

Analytical data for selenium were published earlier by USDA (1992) and were determined by the modified selenium hydride and fluorometric methods. Selenium values for foods analyzed between 1998 and 2008 for NFNAP are determined by either the modified selenium hydride (AOAC 986.15) or stable isotope dilution gas chromatography-mass spectrometry (Reamer and Veillon, 1981) methods. The selenium content of plants, in particular cereal grains, is strongly influenced by the quantity of biologically available selenium in the soil in which the plants grow, that is, by their geographical origin (Kubota and Allaway, 1972). The values given are national averages and should be used with caution when levels of selenium in locally grown foods are of interest or concern.

Beginning with SR19 (2006), values for fluoride, previously released in the USDA National Fluoride Database of Selected Beverages and Foods, Release 2 (USDA, 2005), are included in SR. Other analyzed values in the Fluoride Database, including regional values, are not included in SR. Samples are analyzed using a fluoride ion-specific electrode, direct read method (VanWinkle, 1995) for clear liquids and a micro-diffusion method (VanWinkle, 1995) for other food samples. As with selenium, the